

Exp. No.: 4	Determining the Tool Center Point
Date:	

Aim:

To determine the Tool Center Point (TCP) using automatic calibration in RT Toolbox3.

Procedure:

- Power On and Initialize the Robot
- Command the robot to move to its designated home position.
- Attach a specific tool for to the robot's end-effector for determining the TCP.
- Orient the tool to ensure correct positioning with a workpiece.
- Select Tool Automatic Calculation in Robot name>Tool>Tool Automatic Calculation> Auxiliary point> 1 Point and record the position using Teach Selection Line.
- Repeat the two above steps in two additional orientations for the tool.
- Note the calculated tool coordinate and Presumed Error.
- Rotate the tool about the three axes at the calibrated point to visually verify the calibration.
- Increase the number of points for Teach Selection Line for decreasing the error and write into the controller.
- Send the robot back to its home position once the calibration is finished.

Teach Pendant Mode:

Enable TB enable switch

Pull the lever on the backside of the pendant

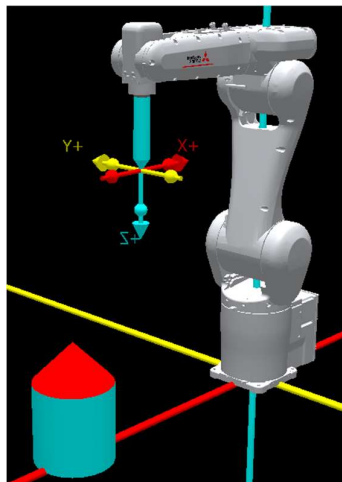
Change Override to <20

Turn on Servos

Orient tool > 1 Point; Orient tool > 2 Point; Orient tool > 3 Point

Perform Tool Automatic Calculation and record error

Simulation:



Hand details

Base = 20,20,150 position =0,0,0

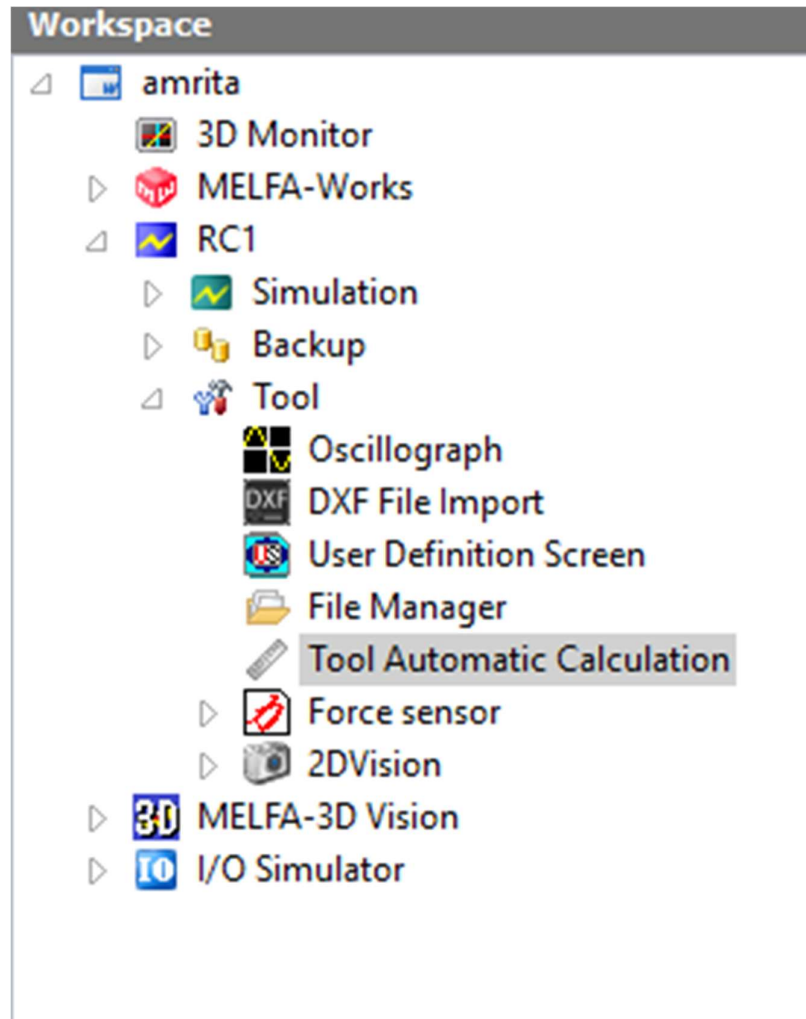
Cone = 3,20,25 position = 0,0,150

Tip =0,0,175 position=0,0,175

Base cylinder and cone details

Cylinder=100,100,200 position =700,0,0

Cone=1,100,100 position =700,0,200



Tool automatic calculation

Capture multiple points in tool automatic calculation to get tool center point for the pen hand and object kept in front

Observation:

- Correlate the error with the deviation in the Tool Center Point and verify that the error decreases by using more points for calibration.

Results & Inference:

RT ToolBox3 - (Tool Automatic Calculation)

Workspace Home Deliew 3D View View Help

Save Save ks Layout File Manager Save Layout File Save Load File Manager More File Save Robot Plots Pits Manager Robot Pertz Fts Save User Mach. Pns Manager User Mertr. File Settings Saves Stop the Robot Interference Check Detailed Simplified Model Type Change Posture of the Robot Click Move Start FPS Rec. View Angle Projection

Workspace ID Monitor Hand Editor: marker? Tool Automats Calculation

MELFA-Works Simulation Program Spline Parameter Monitor Maintenance Board MELFA-Works Backup Tool Oscillograph DXF File Import User Definition File Manager Tool Automatic Force sensor 2DVision MELFA-3D Vision I/O Simulator

Robot1 1: RV-8CRL-D

Tool2

MEXTL2 23.45 -78.10 112.50 0.00 0.00 0.00

Auxiliary point	X	Y	Z
☒ 1 point	123.456	-45.678	987.654
☒ 2 point	234.567	-56.789	876.543
☒ 3 point	345.678	-67.890	765.432
☒ 4 point	456.789	-78.901	654.321
☐ 5 point	567.890	-89.012	543.210
☐ 6 point	678.901	-90.123	432.109
☒ 7 point	789.012	-01.234	321.098
☒ 8 point	890.123	-12.345	210.987

☒ Uses Calculation Teach Selection Line

Calculated Tool Coordinate

1.02 -0.98 145.33 0.00 0.00 0.00

Presumed Error (mm)

0.75 Error Information Write

Output Search

Ready